Rebecca Shannon

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Education

Doctor of Philosophy in Mechanical Engineering Expected May 2027
Boston University, Boston, MA
Advisor: Dr. Sheryl Grace
GPA: 4.0/4.0

M.S. Mechanical Engineering May 2022
B.S. Mechanical Engineering May 2021
Western New England University, Springfield, MA
GPA: 4.0/4.0, Summa Cum Laude, Honors Program, President's List

Research Experience

National Defense Science and Engineering Graduate Fellow 2024 – Present
Unsteady Fluid Mechanics and Acoustic Lab 2022 – Present
Boston University, Boston, MA
Graduate Research Assistant

- Modify native OpenFOAM solver to model shock-droplet interaction with internal bubble dynamics.
- Determine the role of cavitation on droplet breakup in hypersonic-rain interactions.
- Collaborate with researchers at Embry–Riddle Aeronautical University to validate models.
- Supervise undergraduate researchers using Multi-Component Flow Code.

Computation of Fluid and Thermal Processes Lab 2020 – 2022
Western New England University, Springfield, MA
Master's Thesis: *Analysis of particle attribute of microencapsulated phase change materials using long short-term memory*

- Applied a long short-term memory algorithm to an inverse heat conduction problem involving phase change.
- Estimated the size of microencapsulated phase change materials.

Undergraduate Research Assistant

- Developed a heat transfer model and associated parameter identification algorithms.
- Analyzed data to determine the effects of microencapsulation on phase change materials.

Course Project: Solar Panel Snow Plow 2020
Western New England University, Springfield, MA

- Designed and developed prototype to clear snow off solar panels, placed second at the 2020 PDI Showcase and received Grinspoon Entrepreneurial Concept Award.
- Filed a provisional patent application with investors and professors.

Teaching and Work Experience

Graduate Teaching Assistant 2023 – 2024
Boston University, Boston, MA

- Organized discussion sessions for Fluid Mechanics and Statics courses.
- Hosted office hours to individually help students with homework and projects.
- Created rubrics for homework assignments to help manage undergraduate graders.

Undergraduate Teaching Assistant 2019 – 2021
Western New England University, Springfield, MA

- Aided professor in first-year introductory engineering courses and labs.
- Guided undergraduate students to complete coding and laboratory assignments.
- Graded assignments and lab reports.

Cadet Engineer (Summers) 2017 – 2021
Holyoke Gas and Electric, Holyoke, MA

- Created tagging system to identify and organize all valves in a hydroelectric system.
- Performed building condition assessment and various safety inspections.

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Outreach and Involvement

Student Association of Graduate Engineers (SAGE), Communications Chair 2024 – Present
Boston University, Boston, MA

- Organize events for graduate engineering students such as attending musicals or hockey games.
- Write monthly newsletters informing students of upcoming events.
- Manage the SAGE Instagram account.

Peer Mentor 2021 – 2022
Western New England University, Springfield, MA

- Mentored nine first-year engineering students.
- Met monthly with students to share tips to succeed in school.

Awards and Honor Societies

Best Poster Presentation 2026
Boston University Department of Mechanical Engineering Spring 2026 PhD Poster Session.

Graduate Student Teacher of the Year 2024
Boston University Department of Mechanical Engineering for the 2023-2024 academic year.

College of Engineering Distinguished Fellowship in Energy and Sustainability 2022
Boston University College of Engineering's most prestigious fellowship award, presented based on strength of application materials and experiences relevant to the college's cross-disciplinary strategic strengths.

Student Facilitator 2021
Northeast Cyber Team award for computational research using resources at the Massachusetts Green High Performance Computing Center and collaboration with other student researchers.

Marston Award (WNE) 2021
Co-Recipient: Awarded to the senior students judged to have the most mature understanding of the principles of professional engineering practice.

Tau Beta Pi Honor Society, Treasurer (WNE) 2019 – 2021
National honor society for engineering students demonstrating academic excellence.

Dean's Award for Engineering (WNE) 2018 and 2021
Presented for earning the highest cumulative GPA in the School of Engineering in the first and last years in the program.

Selected Publications and Conference Proceedings

R. C. Shannon, R. Forehand, B. Cavainolo, M. Kinzel, and S. Grace, "Computational Simulations to Investigate the Role of Cavitation in Shock-Drop Breakup," UCAH Forum 2025, Alexandria, VA, 2025.

R. Shannon, K. Nguyen, M. Kinzel, and S. Grace, "The role of bubble dynamics on the development of instabilities on a cylindrical droplet," 2024 Cavitation Symposium, Chania, Greece, 2024.

R. C. Shannon, S. Grace, M. Kinzel, and G. Holt, "Influence of Bubble Dynamics on Kelvin Helmholtz Type Interface Instability," 2024 AIAA SciTech Forum, Orlando, FL, 2024, doi: https://doi.org/10.2514/6.2024-2194

R.W. Forehand, K. C. Nguyen, C. J. Anderson, **R. Shannon**, M. P. Kinzel, and S. M. Grace, "An Assessment of Shock-Drop Interaction Modeling Including Cavitation," *Physics of Fluids* 1 February 2023; 35 (2): 023315. doi: 10.1063/5.0136536.

R. C. Shannon, J. Zhao, Z. Li, and J. Benner, "Sensitivity Study of Mini-Batch Size on a Long Short-Term Memory Network for Predicting Shell-to-Core Ratio of Microencapsulated Phase change Materials," 2022 IEEE International Conference on Prognosis and Health Management (ICPHM), Detroit (Romulus), MI, 2022, pp. 77-81, doi: 10.1109/ICPHM53196.2022.9815640.